

इंटरनेट

मानक

Disclosure to Promote the Right To Information

Whereas the Parliament of India has set out to provide a practical regime of right to information for citizens to secure access to information under the control of public authorities, in order to promote transparency and accountability in the working of every public authority, and whereas the attached publication of the Bureau of Indian Standards is of particular interest to the public, particularly disadvantaged communities and those engaged in the pursuit of education and knowledge, the attached public safety standard is made available to promote the timely dissemination of this information in an accurate manner to the public.

“जानने का अधिकार, जीने का अधिकार”

Mazdoor Kisan Shakti Sangathan

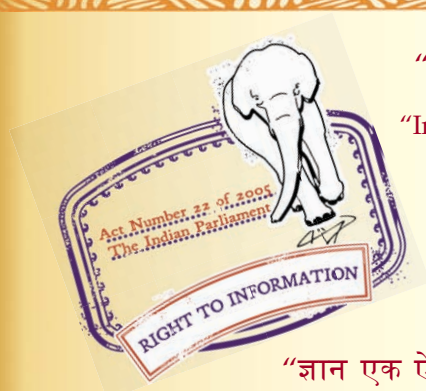
“The Right to Information, The Right to Live”

“पुराने को छोड़ नये के तरफ”

Jawaharlal Nehru

“Step Out From the Old to the New”

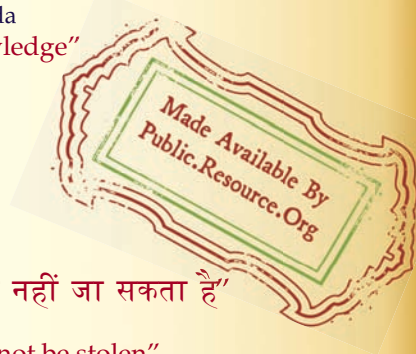
IS 15298-6 (2004): Safety, protective and occupational footwear for professional use, Part 6: Additional specifications for safety footwear [CHD 19: Footwear]



“ज्ञान से एक नये भारत का निर्माण”

Satyanarayan Gangaram Pitroda

“Invent a New India Using Knowledge”



“ज्ञान एक ऐसा खजाना है जो कभी चुराया नहीं जा सकता है”

Bhartrhari—Nitiśatakam

“Knowledge is such a treasure which cannot be stolen”

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IS 15298 (Part 6) : 2004
ISO 8782-6 : 2000

भारतीय मानक

व्यावसायिक उपयोग के लिये सुरक्षित,
संरक्षी और नौकरी पेशा फुटवियर

भाग 6 सुरक्षित फुटवियर की अतिरिक्त विशिष्टियाँ

Indian Standard

SAFETY, PROTECTIVE AND OCCUPATIONAL
FOOTWEAR FOR PROFESSIONAL USE

PART 6 ADDITIONAL SPECIFICATIONS FOR SAFETY FOOTWEAR

ICS 13.340.50

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BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG

NEW DELHI 110002

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Price Group 3

NATIONAL FOREWORD

This Indian Standard (Part 6) which is identical with ISO 8782-6 : 2000 'Safety, protective and occupational footwear for professional use — Part 6 : Additional specifications for safety footwear' issued by the International Organization for Standardization (ISO) was adopted by the Bureau of Indian Standards on the recommendations of the Footwear Sectional Committee and approval of the Chemical Division Council.

The text of ISO Standard has been proposed to be approved as suitable for publication as an Indian Standard without deviations. Certain conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
- b) Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

CROSS REFERENCES

In this adopted standard reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their places are listed below along with their degree of equivalence for the editions indicated:

<i>International Standard</i>	<i>Corresponding Indian Standard</i>	<i>Degree of Equivalence</i>
ISO 8782-1 : 1998	IS 15298 (Part 1) : 2002 Safety, protective and occupational footwear for professional use: Part 1 Requirements and test methods	Identical
ISO 8782-2 : 1998	IS 15298 (Part 2) : 2002 Safety, protective and occupational footwear for professional use: Part 2 Specification for safety footwear	do
ISO 8782-5 : 2000	IS 15298 (Part 5) : 2004 Safety, protective and occupational footwear for professional use: Part 5 Additional requirements and test methods	do

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Introduction

This part of ISO 8782 extends the scope of ISO 8782-2 by specifying further properties which may be needed in safety footwear for professional use. It refers to ISO 8782-2 for the basic requirements which are mandatory for all items of safety footwear and, if desired, allows other combinations of properties to be specified by the selection of appropriate optional requirements from Table 3 of ISO 8782-2:1998.

Indian Standard
**SAFETY, PROTECTIVE AND OCCUPATIONAL
FOOTWEAR FOR PROFESSIONAL USE**
PART 6 ADDITIONAL SPECIFICATIONS FOR SAFETY FOOTWEAR

1 Scope

This part of ISO 8782 specifies, by reference to ISO 8782-1 and ISO 8782-5, additional requirements for safety footwear, relating to water resistance, protection against cutting by hand-held chain saws, resistance to fire fighting hazards (footwear for fire fighters), metatarsal protection and cut resistance.

Basic and further optional requirements for safety footwear are given in ISO 8782-2.

2 Normative references

The following normative documents contain provisions which, through reference in this text, constitute provisions of this part of ISO 8782. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. However, parties to agreements based on this part of ISO 8782 are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below. For undated references, the latest edition of the normative document referred to applies. Members of ISO and IEC maintain registers of currently valid International Standards.

ISO 8782-1:1998, *Safety, protective and occupational footwear for professional use — Part 1: Requirements and test methods*.

ISO 8782-2:1998, *Safety, protective and occupational footwear for professional use — Part 2: Specification for safety footwear*.

ISO 8782-5:1999, *Safety, protective and occupational footwear for professional use — Part 5: Additional requirements and test methods*.

3 Terms and definitions

For the purposes of this part of ISO 8782, the terms and definitions given in ISO 8782-1 and ISO 8782-2 apply.

4 Classification

Footwear shall be classified in accordance with Table 1 of ISO 8782-2:1998.

5 Safety footwear with resistance to water

5.1 Performance

Footwear shall comply with the basic requirements specified in Table 2 of ISO 8782-2:1998 and with the requirement given in Table 1 of this part of ISO 8782.

Table 1 — Safety footwear with resistance to water

Requirement	Appropriate clause in ISO 8782-5:1999	Classification I ^a
Resistance to water	4.2	X
^a X means that the safety footwear shall comply with the requirement.		

5.2 Marking

Each item of safety footwear shall be clearly and permanently marked, for example by embossing or branding, as specified in clause 6 of ISO 8782-2:1998, and with the symbol WR.

6 Safety footwear with protection against cutting by hand-held chain saw

6.1 Performance

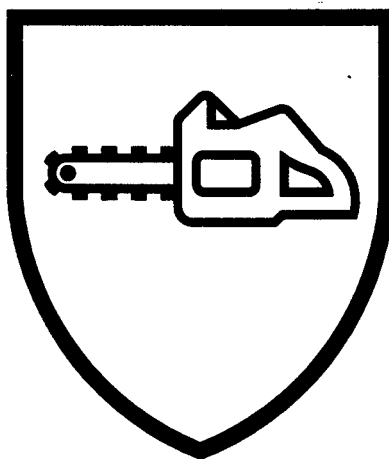
Four classes of footwear are specified according to the different needs of the workplace. Class 0 applies only to classification I footwear and has been introduced on an interim basis to allow manufacturers time to develop footwear of this classification capable of meeting the more rigorous requirements specified for classes 1, 2 or 3. Consequently class 0 will cease to be valid from 1999-12-31.

Footwear shall comply with the basic requirements specified in Table 2 of ISO 8782-2:1998, except where otherwise indicated, and with the requirements given in Table 2 of this part of ISO 8782.

6.2 Marking

Each item of safety footwear shall be clearly and permanently marked, for example by embossing or branding, as specified in clause 6 of ISO 8782-2:1998.

In addition, the pictogram shown in Figure 1, together with the class of protection provided (class 0, class 1, class 2 or class 3), shall be given on a label of dimensions at least 30 mm × 30 mm attached in a visible position on the outside of the footwear.



X = Class of protection

Figure 1 — Pictogram indicating protection against chain saw cutting

Table 2 — Safety footwear with protection against cutting by hand-held chain saw

Requirement	Appropriate clause in		Classification ^a		Indication of change to basic requirements
	ISO 8782-1:1998	ISO 8782-5:1999	I	II	
Cleated outsole	4.8.1.1		X	X	Become mandatory requirements
	4.8.1.2		X	X	
	4.8.1.3		X	X	
Design		4.3.1	X	X	Limitation on designs permitted in ISO 8782-2:1998
Construction		4.3.2	X	X	
Resistance to chain saw cutting:		4.3.3			
class 0			X		
class 1			X	X	
class 2			X	X	
class 3			X	X	
Flex resistance of metal penetration-resistant insert		4.3.4	O	O	
^a X means that the safety footwear shall comply with the requirement. O means that the safety footwear shall comply with the requirement if the component part is present.					

6.3 Information to be supplied by the manufacturer

Each pair of footwear shall be supplied with a leaflet containing the substance of the following wording.

No personal protective equipment can ensure 100 % protection against cutting from a hand-held chain saw. Nevertheless, experience has shown that it is possible to design equipment which offers a certain degree of protection. Different functional principles which may be applied in order to give protection include:

- chain slipping on contact, such that it does not cut the material;
- clogging fibres drawn by the chain into the drive sprocket to block chain movement;
- chain braking by use of fibres with a high resistance to cutting which absorb rotational energy, thereby reducing chain speed.

Often more than one principle is applied.

Four classes of footwear are available, denoting different levels of chain saw resistance. At the present time, only classification II footwear can achieve the highest level. However, in particular working environments and conditions, classification I footwear may be preferred for ergonomic reasons.

It should be noted that the protective area extends to at least 195 mm above the insole in order to ensure that there is an overlap between the footwear and trousers and is not related to the size of the footwear.

7 Safety footwear with resistance to fire fighting hazards

7.1 Performance

Footwear shall comply with the basic requirements specified in Table 2 of ISO 8782-2:1998, except where otherwise indicated, and with the requirements given in Table 3 of this part of ISO 8782.

Table 3 — Safety footwear with resistance to fire fighting hazards

Requirement	Appropriate clause in		Classification ^a		Indication of change to basic requirements
	ISO 8782-1:1998	ISO 8782-5:1999	I	II	
Water penetration and water absorption	4.4.5		X		Becomes mandatory requirement
Cleated outsole	4.8.1.1		X	X	Become mandatory requirements
	4.8.1.2		X	X	
	4.8.1.3		X	X	
Resistance to water		4.2	X		
Design		4.4.1	X	X	Limitation on designs permitted in ISO 8782:1998
Construction		4.4.2	X		
Flex resistance of metal penetration-resistant insert		4.4.3	0	0	
Outsole		4.4.4.1	X	X	
		4.4.4.2	X	X	
		4.4.4.3	X	X	
		4.4.4.4	X	X	
Thermal behaviour		4.4.5.1	X	X	
		4.4.5.2	X	X	
		4.4.5.3	X	X	
^a X means that the safety footwear shall comply with the requirement. O means that the safety footwear shall comply with the requirement if the component part is present.					

7.2 Marking

Each item of safety footwear shall be clearly and permanently marked, for example by embossing or branding, as specified in clause 6 of ISO 8782-2:1998.

In addition, the pictogram shown in Figure 2 shall be given on a label of dimensions at least 30 mm × 30 mm, attached in a visible position on the outside of the footwear.

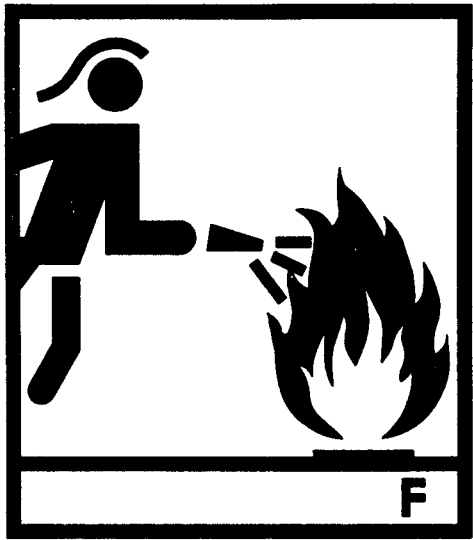


Figure 2 — Pictogram indicating protection against fire fighting hazards

One of the following symbols shall be marked in the bottom right-hand corner of the pictogram:

- F where the requirements specified in 7.1 are met;
- FP where, in addition, the footwear meets the requirements for penetration resistance specified in 4.3.3 of ISO 8782-1:1998;
- FA where, in addition, the footwear meets the requirements for antistatic properties specified in 4.3.4.2 of ISO 8782-1:1998;
- FPA where, in addition, the footwear meets the requirements for both penetration resistance and antistatic properties specified in ISO 8782-1:1998.

8 Safety footwear with metatarsal protection

8.1 Performance

Footwear shall comply with the basic requirements specified in Table 2 of ISO 8782-1:1998 and with the requirements given in Table 4 of this part of ISO 8782.

Table 4 — Safety footwear with metatarsal protection

Requirement	Appropriate clause in ISO 8782-5:1999	Classification ^a	
		I	II
Construction	4.5.1.1	X	X
	4.5.1.2	X	X
	4.5.1.3	X	X
Impact resistance of metatarsal protective device	4.5.2	X	X
^a X means that the safety footwear shall comply with the requirement.			

8.2 Marking

Each item of safety footwear shall be clearly and permanently marked, for example by embossing or branding, as specified in clause 6 of ISO 8782-2:1998, and with the symbol M.

9 Safety footwear with cut resistance

9.1 Performance

Footwear shall comply with the basic requirements specified in Table 2 of ISO 8782-2:1998, except where otherwise indicated, and with the requirements given in Table 5 of this part of ISO 8782.

Table 5 — Safety footwear with cut resistance

Requirement	Appropriate clause in		Classification ^a		Indication of change to basic requirements
	ISO 8782-1:1998	ISO 8782-5:1999	I	II	
Penetration resistance	4.3.3		X	X	Becomes mandatory requirement
Design		4.6.1	X	X	Limitation on designs permitted in ISO 8782:1998
Construction		4.6.2	X	X	
Resistance to cutting		4.6.3	X	X	
^a X means that the safety footwear shall comply with the requirement.					

9.2 Marking

Each item of safety footwear shall be clearly and permanently marked, for example by embossing or branding, as specified in clause 6 of ISO 8782-2:1998, and with the symbol CR.

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This Indian Standard has been developed from Doc : No. CHD 19 (1115).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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**AMENDMENT NO. 1 SEPTEMBER 2005
TO**

**IS 15298 (PART 6) : 2004/ISO 8782-6 : 2000 SAFETY,
PROTECTIVE AND OCCUPATIONAL FOOTWEAR FOR
PROFESSIONAL USE**

PART 6 ADDITIONAL SPECIFICATIONS FOR SAFETY FOOTWEAR

(*Second cover page, para 3*) — Insert the following para after para 3:

As decided by the Committee, additional requirements of marking are given in National Annex A. These additional requirements are part of the standard.

(*Page 6, National Annex*) — Insert the following Annex A at the end of the text:

‘NATIONAL ANNEX A

A-1 BIS CERTIFICATION MARKING

A-1.1 The product may also be marked with the Standard Mark.

A-1.1.1 The use of the Standard Mark is governed by the provisions of *Bureau of Indian Standards Act, 1986* and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.’